PRODUCTIVITY + ON-DEVICE + GENAI

CodeSentinel

Decision-assist layer for repo operations

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Problem Statement

Why This Theme

Engineering teams lose speed due to **context switching** and **ownership confusion**.
CodeSentinel uses local durable memory and GenAI summaries to make decisions faster.

Problem We Solve

Critical repo context scattered across **GitHub, Slack, Jira, CVE feeds**
Causes slow reviews, delayed security fixes, unclear accountability

Why It Matters

Ownership delays increase **MTTR**, block releases, extend security risk windows
Faster clarity = faster **safe merges** and **incident response**

Who Faces It (Persona)

- Tech Lead
- On-call Dev
- PR Reviewer

What Makes It Real

Not just a bot—full workflow:

```
register → CVE heartbeat → blast radius + owner clues
→ PR premortem → auto-fix draft PR → approval loop
→ weekly report
```

Current Solution & Gaps

Today's Reality

GitHub Alerts | Dependabot | Snyk | Slack Threads | Git Blame

Gap: Data exists, but **action context is fragmented**

What's Missing / Broken

Missing a single system combining **risk severity + blast radius + owner + history**.

Broken handoff: Alerts fire, teams still ask **"Who owns this? How risky? Merge now?"**

OPPORTUNITY

Build a **workflow-native control layer** for engineering decisions.

- Premortem on PRs
- Owner Hints
- Auto-fix Draft PRs
- Approval Workflow
- Weekly Rot Visibility

Faster **time-to-decision** & lower **security-response latency**

Our Solution

Decision-Assist Layer

Connects **security risk**, **code ownership**, and **PR guidance** in one flow.

What We Built

CLI + Server + Integrations + Dashboard + Durable Memory

Slack | GitHub | Twilio | Node.js

Don't add another passive dashboard.
Convert scattered signals into workflow-time decisions.

Durable Memory

Dependency ledger, blast radius map, ownership clues, decision history

Ownership Insights

Reduce "who should handle this?" delays with archaeology clues

PR Premortem

Proactive risk review cards for safer merge decisions

Auto-Fix Draft PR

Critical CVE auto-fix flow with human approval gate

Live Canvas & Weekly Digest

Ongoing visibility into repo health and team decisions

Demo / Product Walkthrough

1. Register
Slack: /sentinel register {repo}

2. Detect
Canvas populates dependencies, blast radius, ownership

3. Alert & Guide
Slack Alert: Critical CVE heartbeat | GitHub: Premortem Card

4. Fix & Approve
GitHub: Draft auto-fix PR | WhatsApp: YES/NO approval

Operational Workflow, Not Static Analyzer

Interacts via **Slack** (commands), **GitHub** (reviews), **WhatsApp** (approvals), **Canvas** (status).

Key Moments

ALERT TO ACTION
Critical CVE → Draft fix PR in minutes

ZERO COORDINATION DELAY
Ownership clues + blast radius at decision time

PROACTIVE REVIEW
Premortem before merge, not post-incident

HUMAN CONTROL
Automated suggestions + explicit approval gate

Tech & Architecture

System Flow

Input
Webhooks / User Command

→

Router
Event Dispatcher

Modules
cve-sweep / git-archaeologist / pr-premortem

→

Memory
Durable Repo Memory

Output
Slack Alert / PR Comment / Draft PR / Report

Core Principle: Event-driven Node server routing tasks to specialized modules, all reading/writing shared Durable Memory.

Tech Stack

- **Runtime:**Node.js (ESM), Python
- **Interface:**CLI, Slack, Canvas
- **APIs:**GitHub, Slack, Twilio, NVD
- **Storage:**File-backed Memory

`openclaw` `anthropic-ai/sdk` `octokit/graphql` `slack/webhook` `clack/prompts`

GenAI Powered

AI Usage

- **Model:**OpenRouter
- **Layer:**Centralized Client Module
- **Role:**Raw signals → Actionable context (Not blind merges)

Impact & Use Cases

Reduced Time-to-Decision

X10

Faster incident response & safer PR decisions by providing **Risk + Owner + Action Path** in one place.

CRITICAL CVE

Alert triggers, blast radius shown, draft fix PR created, approver confirms quickly.

NEW ENGINEER

/why {file} surfaces ownership clues + history context, reducing tribal knowledge reliance.

RISKY PR

Premortem card highlights likely breakpoints and missing reviewers before merge.

MONOREPO ROT

Weekly rot report exposes unresolved risk/ownership hotspots for leadership follow-up.

Scale Potential

Automated | API-driven | Memory-backed | Consistent Risk Triage

Differentiation

MARKET SHIFT

- ~~Point Solutions~~ → Decision Workflow
- ~~Passive Reporting~~ → Time-of-Decision Automation
- ~~Resetting Context~~ → Durable Memory

We connect Detection → Context → Action → Approval inside tools teams already use.

Compounding Memory Graph

01

File-level ownership, risk history, and module passports become a **proprietary org knowledge layer** that strengthens with every event.

Workflow Embedding + Trust Loop

02

Deep integration into **Slack + GitHub + Approval** channels creates behavioral lock-in; relied upon for real security decisions.

Demo & Resources

GitHub Repo

https://github.com/zenodefault/codesentinel

Live Demo

https://drive.google.com/file/d/1QwqJI9u10inqc_LCiV94QfAQjePB6oqw/view?usp=drivesdk

Documentation

https://docs.google.com/document/d/1bXlZdioGcbF3ev51J_KF7Ookh83HCoCF/edit?usp=drive_link&ouid=117547128866674394060&rtpof=true&sd=true

Beyond The Ask

Repo Intelligence Layer

Not just solving alert noise—building **institutional memory** that improves every week. Reducing reliance on tribal knowledge.

Human-in-the-loop
Balance automation with control

Weekly Rot Intel
Proactive engineering hygiene

Onboarding
Historical context + ownership

Future-Ready Extensions

- Jira auto-ticketing from risk paths
- Release risk score per PR/Sprint
- Org-level reliability cockpit

From "Who owns this?" to Clear Action in Minutes.